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# Artificial yet credible: How virtual influencers overcome authenticity challenges

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ABSTRACT

Virtual influencers (VIs) present a paradox as they can be perceived as highly credible yet fundamentally inauthentic. Drawing on cognitive dissonance theory, this research examines how consumers resolve this conflict. Across two experiments, we show that credibility functions as a psychological repair mechanism, offsetting authenticity deficits and enabling persuasion.

We advance authenticity theory by revealing its divergent dimensionality for VIs. Consumers decouple the messenger from the message, allowing high influencer authenticity (true to character) to coexist with low claim-based authenticity (endorsed-value, product, message authenticity). We also identify an authenticity penalty, where high-realism VIs are penalised on claim-based authenticity for violating human-like expectations. We further show that trustworthiness and expertise repair claim-based and influencer authenticity, while attractiveness repairs only endorsed-value authenticity yet reduces message authenticity. The efficacy of this mechanism depends on endorsed value type and perceived product value. This research resolves a foundational paradox in digital influence of artificial agents.

## 1. Introduction

Influencer marketing has reshaped brand communication, providing brands with innovative ways to reach and engage consumers. Influencers build trust through personal storytelling and often achieve greater credibility than traditional advertising or celebrity endorsements (Zhou et al., 2021). The global influencer industry is projected to reach $84.89 billion by 2028, with 75 % of firms already collaborating with influencers and 43 % planning further investments (Influencer Marketing Hub, 2024). However, rapid industry growth has raised concerns about authenticity. Commercial pressures risk diluting the genuine connections between influencers and consumers, potentially threatening consumer engagement and brand equity (Cheah et al., 2024; Audrezet et al., 2020).

The rise of virtual influencers (VIs) has intensified challenges around authenticity. These AI-driven personas, managed by individuals, agencies, or brands (Zhou et al., 2024), are increasingly effective at shaping brand perceptions (Lou et al., 2023), encouraging electronic word-of-mouth (eWOM) (Sands et al., 2022), and driving purchase (Thomas & Fowler, 2021), mirroring the impact of human influencers (HIs). Yet VIs fundamentally challenge traditional notions of authenticity (Koles et al., 2024). Rather than being a single, uniform concept, authenticity consists of multiple dimensions that may react differently to virtual entities (Nunes et al., 2021). Consumers might find a VI's persona consistent and believable within its narrative while doubting the authenticity of its commercial endorsements. This blurring of the real and the algorithmic creates psychological ambiguity and complicates consumer perceptions.

Research on VIs reveals a paradox. VIs, despite lacking the lived experience that underpins HI authenticity (Dondapati & Dehury, 2024), can be perceived as highly credible and believable (Sands et al., 2022), even in some cases outperforming HIs (da Silva Oliveira & Chimenti, 2021). Existing authenticity models, developed primarily for HIs (e.g., Audrezet et al., 2020; Liao & Chen, 2024), do not fully explain how consumers reconcile VIs' credibility with their perceived lack of authenticity. This gap leaves critical theoretical questions unresolved. How do consumers resolve the tension between credibility and authenticity of VIs, and what mechanisms drive persuasion in this context?

Clarifying the nature of authenticity in VI marketing is increasingly urgent. VIs are positioned as valuable brand ambassadors, with the potential to accelerate the adoption of emerging technologies such as

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virtual reality and artificial intelligence (Audrezet et al., 2025). Yet, their artificiality presents a challenge for brands seeking authentic connections. Addressing this authenticity deficit is significant, as consumer scepticism can reduce campaign effectiveness, such as endorsed-value adoption, purchase intention, and eWOM intention. Understanding how consumers navigate these tensions can inform campaign design and foster favourable outcomes.

In addition, for marketers, the choice between HIs and VIs is increasingly strategic. Brands weigh the control, scalability, and novelty of VIs against the lived experience and emotional resonance offered by HIs. This decision often depends on the nature of products and campaign objectives. Utilitarian products that require expertise and consistency may benefit more from VIs, whereas hedonic or identity-expressive products usually require the emotional authenticity and social connection of HIs. While this research focuses on VIs, understanding how types of endorsed-value and levels of consumer perceived value shape influencer effectiveness is essential for optimising strategy. This perspective recognises the ongoing debate and highlights the importance of aligning influencer selection with campaign goals and value propositions.

This research is among the first to empirically examine whether endorser credibility can compensate for low authenticity perceptions in VI marketing. Drawing on cognitive dissonance theory (Festinger, 1957), we propose that credibility provides consumers with rational justifications to overlook authenticity deficits, thereby reducing cognitive dissonance and promopting positive behavioural outcomes. Through two experiments, we investigate how influencer type (HIs, high-realism VIs, and low-realism VIs) affects endorsed-value adoption, purchase intention, and eWOM intention. We examine serial mediation via endorser credibility attributes (attractiveness, expertise, trustworthiness) and distinct authenticity dimensions (influencer, message, product, endorsed-value). We further explore how endorsed-value type (personal, social, original) and perceived value moderate these effects.

Our findings make several contributions. First, we address the authenticity-credibility paradox in VI marketing. While traditional linear models assume credibility is built on authenticity (Srivastava et al., 2020; Wang & Weng, 2024), our results show that high-realism VIs can be perceived as highly credible despite limited authenticity. This finding challenges the assumption that believable endorsers should first be “real.” Unlike both linear and parallel models (Cheah et al., 2024; Kim & Wang, 2024), we show that VIs’ superior credibility can offset authenticity deficits, serving as a dissonance-reduction mechanism that drives positive consumer responses.

Second, we advance the conceptualisation of authenticity in influencer marketing by empirically distinguishing its dimensions (e.g., Cheah et al., 2024; Kim & Wang, 2024; Wang & Weng, 2024). Moving beyond a monolithic view, we show that influencer, message, product, and endorsed-value authenticity can diverge. VIs could be seen as authentic in influencer persona but inauthentic in message, product, or endorsed-value. This psychological separation nuances the authenticity management and explains how a fake entity can still be regarded as an effective endorser.

Third, we contribute to anthropomorphism theory by showing that increased VI realism produces both credibility rewards and authenticity penalties. High-realism VIs are often perceived as more credible but less authentic than low-realism VIs. Their pronounced human-like features heighten consumer expectations for human qualities, therefore, any perceived artificiality leads to a stronger sense of violated expectations and, consequently, an authenticity penalty. Moreover, in challenging recent literature on HIs’ superiority (Lou et al., 2023; Lee et al., 2024), we find that high-realism VIs can achieve higher credibility than HIs, reflecting that both technological advances and shifting consumer perceptions are redefining digital influence.

Finally, we extend the VI marketing literature by highlighting the importance of value proposition for repairing authenticity and enhancing persuasion. The effectivenss of credibility in compensating for authenticity deficits depends on endorsed-value type, particulalrly personal and social values, strengthening the repair process and lead to distinct behavioural outcomes. Moreover, the persuasiveness of the authenticity-credibility trade-off depends on consumers’ perceived value of the product, and is most impactful when products are perceived as meaningful for self-expression, social signalling, or emotional attachment.

By clarifying these mechanisms, our research provides actionable guidance for brands employing VI marketing. Leveraging endorser credibility can mitigate consumer scepticism and improve campaign effectiveness, ultimately driving more positive consumer behaviours.

## 2. Theoretical background

### 2.1. Authenticity in influencer marketing: A multifaceted construct

Authenticity is a foundational concept in marketing, shaping perceptions of what is genuine and true (Becker et al., 2019). In consumer behaviour, authenticity is widely recognised as multidimensional, comprising facets such as accuracy, connectedness, integrity, legitimacy, originality, and proficiency (Nunes et al., 2021). Consumers assess these dimensions contextually, often prioritising certain aspects depending on the brand or message. In advertising, authenticity has been classified into pure, approximate, and moral forms (Beverland et al., 2008), while emphasising the importance of preserving brand essence, heritage, and credible storytelling (Becker et al., 2019).

In influencer marketing, authenticity is central to persuasive impact (Audrezet et al., 2020). Influencer authenticity reflects perceived transparency, genuineness, and alignment with personal values, rather than endorsements driven simply by financial motives (Lee & Eastin, 2021). Influencers cultivate authenticity by sharing personal experiences (Gannon & Prothero, 2016), creating original content (Duffy & Wissinger, 2017), and engaging directly with followers (Li & Ma, 2024).

As consumer scepticism increases, brands should manage and communicate authenticity more strategically in influencer partnerships (Zniva et al., 2023). Beyond influencer authenticity, other relevant dimensions include message, product, and endorsed-value authenticity. Message authenticity concerns the transparency and sincerity of content (Audrezet et al., 2020). Product authenticity reflects the alignment between the product, influencer, and consumer expectations (De Veirman et al., 2017). Endorsed-value authenticity involves the perceived genuineness of values, lifestyles, or benefits associated with the product as well as the fit between the influencer’s perceived values, the brand’s values, and those embodied by the product (Audrezet et al., 2020).

Despite its importance, most recent research has focused narrowly on influencer authenticity, leaving gaps in understanding how authenticity operates across domains. These issues are exacerbated by VIs, whose artificial nature further complicates perceptions of authenticity (Lou et al., 2023). As summarised in Table 1, prior studies have primarily examined HI authenticity or treated authenticity as a unidimensional construct. Research on VI authenticity has been qualitative or has not integrated multiple authenticity dimensions with endorser credibility in a unified framework. Extending this literature, our research directly compares HIs with both high- and low-realism VIs, operationalises four distinct authenticity dimensions, and embeds these constructs within a cognitive dissonance theory framework (Festinger, 1957) to show how credibility can compensate for authenticity deficits.

### 2.2. Anthropomorphism in VI marketing

Anthropomorphism, the tendency to attribute humanlike qualities or behaviours to non-human agents, provides a critical lens to understand how consumers respond to VIs (Epley et al., 2007). The realism of a VI shapes the degree of perceived anthropomorphism. According to the computers are social actors (CASA) paradigm, people instinctively apply social rules to technology (Nass & Moon, 2000), treating highly realistic VIs as social actors, and expecting humanlike warmth, competence, and

**Table 1**
Summary of authenticity domains studied in influencer marketing research.

| Study | Influencer Type | Aim | Authenticity Domain(s) | Theory | Method | Participants/Unit of Analysis | Main Findings |
|---|---|---|---|---|---|---|---|
| Audrezet et al. (2020) | HIs | Examine how influencers manage authenticity in brand partnerships | Influencer authenticity | Social influence and social learning theory | Non-participative observations and semi-directive interviews | N = 15 French influencers; interviews plus observed content; gender and age varied. | HIs manage passionate and transparent authenticity to balance commercial goals with audience expectations. |
| Kapitan et al. (2022) | HIs | Assess the impact of endorser type and authenticity on willingness to pay | Influencer authenticity | Attribution theory | 4 experiments | N = 745 (across four studies); US participants from Amazon Mechanical Turk (MTurk); 48 % female; $M_{age}$ = 36.8. | Content creators who retain control over endorsements are perceived as more authentic and increase willingness to pay. |
| Zniva et al. (2023) | HIs | Identify antecedents and outcomes of influencer authenticity | Influencer authenticity, brand authenticity | Stimulus-organism-response model | 2 experiments | N = 384; Austrian university students; 61 % female; $M_{age}$ = 22.5. | Influencer uniqueness and consistency enhance perceived authenticity. Influencer authenticity drives purchase intention via brand authenticity and attitudes. |
| Ren et al. (2023) | HIs | Investigate the roles of authenticity and regulatory focus in influencer effects | Influencer authenticity | Parasocial relationship theory | 3 experiments | N = 621 (across three studies); US participants from MTurk; 51 % female; $M_{age}$ = 37.1. | Non-celebrity HIs generate higher purchase intentions; effects are mediated by perceived authenticity and regulatory focus. |
| Andonopoulos et al. (2023) | HIs | Explore how different levels of personal authenticity affect consumer responses | Influencer authenticity | Social learning, transformational leadership, and signalling theory | 4 experiments | N = 890 (across four studies); US & UK participants from Prolific and university pools; 58 % female; $M_{age}$ = 29.4. | Higher perceived authenticity generally boosts behavioural intentions, but even inauthentic HIs can inspire, particularly for hedonic offerings. |
| Koles et al. (2024) | VIs | Explore forms and meanings of authenticity in VIs | Influencer authenticity | Entity-Referent Correspondence Framework of Authenticity | 11 expert interviews and 64 consumer interviews | N = 11 industry professionals and N = 64 consumers (UK & US); 55 % female, age 19–58. | VIs exhibit "true-to-ideal" authenticity but lack "true-to-fact" and "true-to-self." Authenticity may be transferred from human creators. |
| Dondapati & Dehury (2024) | VIs vs. HIs | Examine roles of homophily, parasocial ties, and authenticity in purchase intent | Influencer authenticity | Parasocial relationship theory | 1 experiment | N = 355; Indian online panel; 45 % female; $M_{age}$ = 28.5. | Homophily and parasocial ties mediate the link between influence type and purchase intention; effects are stronger for HIs when authenticity is high. |
| Wang & Weng (2024) | HIs | Analyse effects of authenticity dimensions on credibility and WOM | Influencer authenticity | Social exchange theory | Survey | N = 650; Taiwanese social media users; 53 % female; $M_{age}$ = 31.7. | Distinct authenticity facets (originality, naturality, continuity) differentially enhance influencer credibility and WOM. |
| Li & Ma (2024) | VIs | Test effects of VI humanlikeness on authenticity and advertising outcomes | Influencer authenticity | Computers Are Social Actors (CASA) paradigm | 2 experiments | N = 512 (across two studies); US online panel; 52 % female; $M_{age}$ = 35.2. | More humanlike VIs are perceived as more authentic, which improves ad attitudes and purchase intentions. |
| **Current research** | HIs vs. VIs (high- vs. low-realism VIs) | Explain how credibility can mitigate low-authenticity effects in VI marketing | Endorsed-value authenticity, influencer authenticity, message authenticity, product authenticity | Cognitive dissonance theory (CDT) | 2 experiments | N = 1503 (across two studies); Western participants from 9 countries from Prolific; 52 % female, $M_{age}$ = 36.5. | Directly compares HIs with both high- and low-realism VIs; Operationalises four distinct authenticity dimensions and integrates them with endorser credibility in a CDT framework; Shows that credibility can reduce dissonance arising from authenticity deficits, |

(*continued on next page*)

**Table 1** (*continued*)

| Study | Influencer Type | Aim | Authenticity Domain(s) | Theory | Method | Participants/Unit of Analysis | Main Findings |
|---|---|---|---|---|---|---|---|
| | | | | | | | enabling positive behavioural responses by focusing on expertise and trustworthiness. |

*Notes*: Studies were selected based on their explicit focus on authenticity in influencer contexts and inclusion of either HIs or VIs.

authenticity (Ma & Li, 2024).

However, increasing VI realism introduces the risk of the "uncanny valley" effect, where agents that appear almost human can elicit discomfort or unease (Lou et al., 2023). According to the CASA paradigm, high-realism VIs trigger strong social expectations through humanlike cues. When these cues fail to convincingly replicate human authenticity, they may provoke negative reactions consistent with the uncanny valley phenomenon. This tension is further explained by expectancy violation theory, which posits that failing to meet salient social norms or expectations leads to negative evaluations (Burgoon, 2015). As a result, high-realism VIs may be perceived as more credible due to their human resemblance, but they also face heightened scrutiny over authenticity, creating a psychological conflict for consumers.

Anthropomorphic cues in high-realism VIs trigger cognitive dissonance by prompting consumers to interact with them as if they were human, while remaining aware of their artificial and commercial origins. This tension between human-like engagement and knowledge of artificiality underpins the dissonance addressed in this research. HIs naturally meet anthropomorphic expectations, while high-realism VIs approximate these expectations in appearance and behaviour, and low-realism VIs remain clearly artificial. Thus, realism operationalises anthropomorphic cues and shapes the extent to which consumers apply social expectations, leading to dissonance when authenticity is perceived as lacking.

In our model, anthropomorphism amplifies human-like expectations for influencers. When high realism in VIs intensifies these expectations but authenticity is insufficient, cognitive dissonance theory explains how consumers resolve the resulting discomfort.

### 2.3. Cognitive dissonance theory

Cognitive dissonance theory (Festinger, 1957) provides a valuable framework for examining the paradox between authenticity and credibility in VI marketing. The theory holds that individuals experience psychological discomfort when faced with conflicting beliefs and seek consistency by adjusting attitudes or rationalising choices. In the influencer context, this dynamic becomes salient when consumers, drawn to VIs for recommendation, inspiration, or entertainment, recognise the influencer's artificial and commercial nature. The resulting clash between positive attitudes and knowledge of artificiality creates cognitive dissonance.

While cognitive dissonance is well established in marketing research, such as brand management (e.g., Yakın et al., 2023; Wasaya et al., 2016) and advertising effectiveness (e.g., Spangenberg et al., 2003), its application to VI marketing is limited. This theory helps explain the discomfort consumers feel when they are attracted to highly anthropomorphic VIs yet unsettled by artificiality and commercial intent. Our research extends this theory by investigating how endorser credibility may serve as a mechanism to reduce this dissonance.

Endorser credibility encompasses the qualities of being knowledgeable and reliable (expertise), honest and genuine (trustworthiness), and appealing or relatable (attractiveness) (Lou & Yuan, 2019). Credible HIs are found to enhance brand attitudes and drive purchase intentions (Sokolova & Kefi, 2020; Wang & Weng, 2024). VIs, despite lacking human authenticity, seek to establish credibility by simulating expertise (Miao et al., 2022), leveraging attractiveness (da Silva Oliveira & Chimenti, 2021), and fostering parasocial interaction (Jin & Ryu, 2020). Previous research has largely treated authenticity and credibility as independent constructs (Cheah et al., 2024), or as sequential mediators (Kim & Wang, 2024). Their interactive role in resolving dissonance, particularly for VIs, remains underexplored. Our research addresses this gap by examining whether high endorser credibility can compensate for low authenticity, thereby reducing dissonance and supporting positive consumer responses.

## 3. Hypotheses development

### 3.1. Impact of influencer type and endorsed value type on consumer behaviours

According to cognitive dissonance theory, consumers seek consistency between their beliefs about influencers, endorsed-values, and their own self-concept. When influencers are perceived as congruent with these elements, message internalisation and behavioural intentions become psychologically easier, as no dissonance repair is needed. When congruence is lower, consumers may either reject the message or psychologically resolve the dissonance. These dynamics shape how influencer type and endorsed-value type affect consumer behaviours.

Influencer type can be a key determinant of consumer response. HIs are typically more effective because their lived experiences, personal narratives, and genuine emotions foster trust and relatability (Audrezet et al., 2020; Lou & Yuan, 2019). These qualities drive endorsed-value adoption, purchase intention, and eWOM intention (Lee & Eastin, 2021), capturing a comprehensive view of influencer impact. Endorsed-value adoption reflects the internalisation of the message. Purchase intention captures the likelihood of commercial action. eWOM intention, though often follows accumulated experience and repeated exposure, may capture immediate advocacy and brand sharing after a single exposure (Zhou et al., 2021).

In contrast, VIs, especially those with low-realism, often lack the emotional depth and lived experience needed to build authenticity, leading to greater consumer scepticism (Zhou et al., 2024). Even high-realism VIs, though more humanlike in appearance and behaviour, remain inherently artificial, which can erode trust and limit their impact (Dondapati & Dehury, 2024). Therefore, we hypothesise:

> ***H1**: HIs will have a greater positive effect on consumer behaviours (endorsed-value adoption, purchase intention, and eWOM intention) than both high- and low-realism VIs.*

Consumers are more likely to engage with values that align with their personal beliefs, interests, and identities (Sweeney & Soutar, 2001). Therefore, endorsed-value type also may play a critical role in shaping consumer responses. Influencers typically endorse three types of values, including original, social, and personal values (Zhou et al., 2021). Original values, developed by companies to emphasise a product's benefits or novelty, often emphasise utilitarian benefits such as functionality and performance. Social values, shaped by group norms and societal expectations, and personal values, derived from individual experiences and preferences, more strongly reflect hedonic or symbolic benefits.

Personal values are potentially more effective in evoking positive responses, as they resonate emotionally and psychologically with

consumers, aligning with self-concept and personal goals (Han & Balabanis, 2024). Endorsements centred on personal values are more likely to drive message internalisation, motivate purchase, and inspire eWOM. Social values may influence behaviour when strongly tied to group identity, while original values, focused on product features, may lack the emotional connection needed for stronger engagement. Based on this reasoning, we propose that:

***H2**: Compared to original and social values endorsed by influencers, personal values will have a greater positive effect on consumer behaviours (endorsed-value adoption, purchase intention, and eWOM intention).*

### 3.2. Impact of influencer type and endorsed value type on endorser credibility

Influencer type is expected to directly shape endorser credibility by influencing perception of expertise, trustworthiness, and attractiveness. HIs build credibility through lived experiences, authenticity, and relatability, which foster trust and parasocial relationships (Sokolova & Kefi, 2020). In contrast, high-realism VIs challenge conventional credibility norms by mimicking humanlike traits such as attractiveness, emotional expressions, and nuanced behaviour (Miao et al., 2022). Notably, research indicates that high-realism VIs may even surpass HIs in credibility under conditions where brand safety and consistency are prioritised, as they are immune to scandals and align closely with brand messaging (da Silva Oliveira & Chimenti, 2021). They also tend to outperform low-realism VIs, whose limited anthropomorphic features reduce perceived credibility despite their novelty (Jin & Ryu, 2020; Miao et al., 2022). We thus propose:

***H3**: High-realism VIs will be perceived as more credible than HIs and low-realism VIs in terms of expertise, trustworthiness, and attractiveness.*

Endorsed-value type, reflecting the distinct product benefits and importances promoted by influencers, is less likely to directly affect endorser credibility. Credibility is primarily driven by influencers' perceived realism, professionalism, and relatability rather than by the specific value endorsed (Koles et al., 2024). While value alignment may increase consumer affinity, it does not necessarily elevate perceptions of expertise, trustworthiness, or attractiveness. As such, the following hypothesis is proposed:

***H4**: Endorsed-value type will have no significant direct effect on endorser credibility.*

### 3.3. Authenticity-credibility paradox

Endorser credibility is central to influencer marketing effectiveness, positively shaping purchase and eWOM intentions (Lou & Yuan, 2019; Wang & Weng, 2024; Sokolova & Kefi, 2020). It mediates the relationship between influencer type and consumer responses, fostering message internalisation, purchase intention, and brand advocacy. High-realism VIs often surpass both HIs and low-realism VIs in perceived credibility due to their consistent behaviour, aesthetic appeal, and immunity to scandals, especially in contexts that value consistency and professionalism (Jin & Ryu, 2020). In contrast, low-realism VIs, with visibly artificial designs, typically struggle to establish comparable credibility, limiting their persuasive impact. We therefore propose:

***H5**: Endorser credibility will positively mediate the relationship between influencer type and consumer behaviours (endorsed-value adoption, purchase intention, and eWOM intention).*
***H5a**: High-realism VIs will have stronger indirect positive effects than HIs and low-realism VIs on consumer behaviours through enhanced endorser credibility.*

Authenticity also shapes consumer responses, as consumers increasingly seek genuine, value-driven interactions with influencers (Sokolova & Kefi, 2020; Yuan & Lou, 2020). Perceived authenticity is expected to mediate the impact of influencer type on consumer behaviours across influencer, message, product, and endorsed-value dimensions. When authenticity is perceived as low, it fundamentally constrains persuasion, reducing consumer willingness to internalise messages, consider purchases, or distribute eWOM.

However, high-realism VIs present a paradox. Their humanlike features raise consumer expectations for authenticity, yet their artificiality often leads to a perceived authenticity deficit. This incongruity increases scepticism, as consumers compare high-realism VIs to HIs and find them lacking in lived experience and emotional resonance (Jin & Ryu, 2020). HIs, by contrast, are inherently perceived as more authentic. Low-realism VIs, with their overt artificiality, may evoke lower expectations for authenticity, and provoke less scepticism than high-realism VIs. Therefore, we hypothesise:

***H6**: Perceived authenticity (across influencer, message, product, and endorsed-value dimensions) will negatively mediate the relationship between influencer type and consumer behaviours (endorsed-value adoption, purchase intention, and eWOM intention).*
***H6a**: High-realism VIs will suffer a greater authenticity penalty than HIs and low-realism VIs, manifesting as a stronger negative indirect effect on consumer behaviours via perceived authenticity.*

Cognitive dissonance theory (Festinger, 1957) helps explain how consumers resolve tensions between high credibility and low authenticity in VIs. When confronted with this dissonance, consumers may rationalise their engagement by focusing on influencers' expertise, trustworthiness, or attractive qualities rather than on the lacked authenticity. This cognitive adjustment can restore consistency and enable favourable responses, from endorsed-value adoption to purchase and eWOM. Thus, the interplay between credibility and authenticity is critical. While influencer type shapes both perceptions, high credibility may offset low authenticity, supporting positive behaviours. Thus, the following hypotheses are proposed:

***H7**: Influencer type influences consumer behaviours through a sequential mediation pathway that VIs first determine perception of endorser credibility, which then shapes authenticity judgments (across influencer, message, product, and endorsed-value dimensions) through a psychological repair process, ultimately increasing consumer behaviours (endorsed-value adoption, purchase intention, and eWOM intention).*

### 3.4. Moderating role of endorsed value type

The relationship between endorser credibility and perceived authenticity depends on the nature of the value being endorsed, as different value claims require varying levels of source validation. Drawing on social identity theory (Turner et al., 1979), we posit that social values, which reflect group norms and collective ideals, carry higher social risk and thus require greater credibility from the influencer to legitimate product values and authenticity. In contrast, original values, centred on utilitarian product features, can be verified more objectively and demand less reliance on the influencer's credibility. Personal values, related to self-expression and individuality, fall between these extremes. As a result, influencer credibility is most critical for authenticating social value claims, less so for personal values, and weakest for original values. Therefore, we propose:

***H8**: The repairing effect of endorser credibility on authenticity (endorsed-value authenticity and product authenticity) will be moderated by endorsed-value type.*
***H8a**: This effect will be strongest for social values, followed by personal values, and weakest for original values.*

### 3.5. The moderating role of perceived values

The effect of credibility-restored authenticity on consumer behaviour is contingent on perceived value, which reflects consumers' assessments of product quality, benefits, and the psychological or social relevance of products (Sweeney & Soutar, 2001). Relevant dimensions of perceived value include self-expression, social signalling, and emotional attachment, each reflecting how the product satisfies consumers' need for identity affirmation (Jaiswal et al., 2024), social belonging (Farivar & Wang, 2022), or emotional resonance (Li et al., 2023).

When consumers perceive high value in endorsed products, authenticity is more likely to drive value adoption, purchase, and eWOM (Lou & Yuan, 2019). For instance, self-expressive consumers respond more to authentic endorsements aligning with personal values, while those prioritising social ideals (e.g., sustainability) act when authenticity and value alignment converge (Audrezet et al., 2020). Conversely, when perceived value is low, even authentic endorsements are unlikely to motivate action. Thus, authenticity's positive impact on behaviour intensifies as perceived value increases. Accordingly, we propose:

***H9**: The effect of perceived authenticity (endorsed-value authenticity and product authenticity) on consumer behaviours (endorsed-value adoption, purchase intention, and eWOM intention) will be moderated by perceived values.*

***H9a**: The positive effect will be stronger at higher levels of perceived values.*

## 4. Theoretical framework

Grounded in cognitive dissonance theory (Festinger, 1957), our theoretical framework (see Fig. 1) proposes that VI type (high vs. low-realism) shapes an authenticity-credibility paradox. VIs enhance endorser credibility (trustworthiness, expertise, attractiveness) but simultaneously reduce perceived authenticity across influencer, message, product, and endorsed-value dimensions. Credibility compensates for this authenticity deficit, restoring psychological consistency, and facilitating positive consumer outcomes (endorsed-value adoption, purchase intention, eWOM). This process is further moderated by the type of value endorsed (social, personal, original) and consumers' perceived value of the product.

## 5. Overview of studies

We conducted two experiments across different product categories and influencer types to enhance generalisability and reliability. Study 1 examined whether endorser credibility mitigates the negative effects of low authenticity (endorsed-value authenticity) on consumer behaviour (endorsed-value adoption), and explored how endorsed-value type moderates the relationship between endorser credibility and authenticity. Study 2 extended the analysis by inverstigating the specific dimensions of endorser credibility (expertise, trustworthiness, and attractiveness) and authenticity (product, message, endorsed-value, and influencer). This approach offered a more nuanced understanding of their interplay in shaping consumer behaviours. This study also assessed the moderating roles of endorsed-value type and perceived value in these relationships.

Data collection for Study 1 took place in October 2023, followed by Study 2 in April 2024. To ensure construct validity, we defined all key terms (e.g., HIs, VIs, and endorsed-value types) and provided detailed endorsement contexts to participants during pretests and main studies. Participants were recruited from 9 Western markets, screened as active influencer followers aged 18 to 48, consistent with the core influencer marketing demographic (Influencer Marketing Hub, 2025). All measurement items were randomised to minimise response bias, and demographic data were collected at the end of each study. Attention checks involving a question and time-to-completion filter (less than 2 min) were used in both experiments to ensure data quality by excluding inattentive participants.

## 6. Study 1

Study 1 examines whether endorser credibility and endorsed-value authenticity sequentially mediate the effect of influencer type on endorsed-value adoption, and whether this process is moderated by endorsed-value type.

### 6.1. Method

This study used a 3 (influencer type: HIs, high-realism VIs, low-realism VIs) × 3 (endorsed-value type: original, social, personal) between-subjects design. A total of 920 participants were recruited via Prolific (50.1 % female; 34.0 % aged 29–38, 26.8 % aged 18–28, and

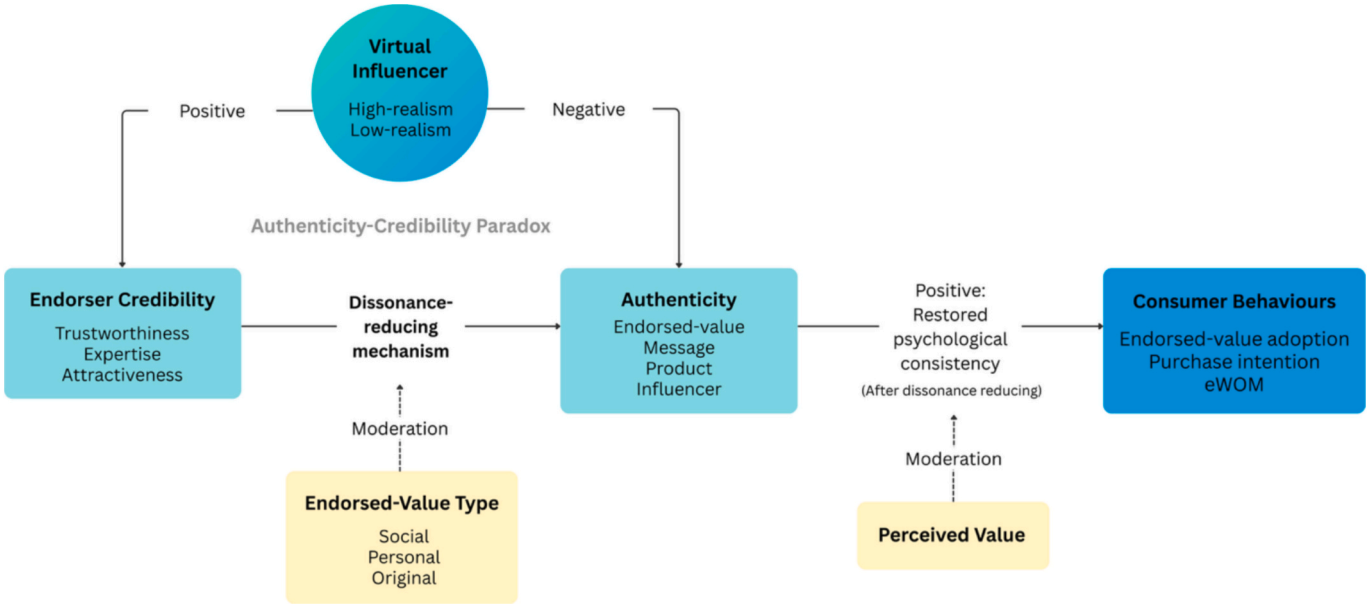



**Fig. 1.** Theoretical framework: Addressing VIs' authenticity-credibility paradox.

19.2 % aged 39–48), resulting in $N = 906$ after removing those who failed attention checks. Each participant viewed a profile (see Appendix A) of a fictitious male influencer, "Taylor," modelled after a real-life VI, Liam Nikuro.

To ensure the validity for VI realism manipulation (high-realism VIs vs. low-realism VIs), we operationalised realism as a categorical variable representing distinct visual styles (hyper-realistic or stylised) prevalent in influencer marketing. This approach aligns with Jin and Ryu (2020) and Zhou et al. (2024), who distinguish between human-like and anime-like VIs. In the HI condition, Taylor was presented as human, with his image generated using the AI platform Generated Photos (https://generated.photos), which produces royalty-free, ethically sourced images of non-existent individuals. This approach ensured ethical integrity by avoiding the use of real influencer's likenesses and pre-existing attitudinal biases. Photo parameters (e.g., face, gender, age, eye and hair colour) were matched to Taylor's profile to maintain consistency. In the high-realism VI condition, the same image was used, but Taylor was described as virtual, reflecting the growing indistinguishability between advanced VIs and humans (Zhou et al., 2024). For the low-realism VI, Taylor was depicted with visibly anime-like features derived from the original image to create a stylised look. A pretest ($N = 256$) confirmed that there were no significant differences in perceived familiarity ($p = 0.54$) or attitudes ($p = 0.22$) across influencer types, minimising confounds.

Participants imagined shopping for a camera and encountered a Facebook post from Taylor endorsing a 4K Digital Camera. The camera category was chosen for its blend of technical attributes (original values), group benefits (social values), and lifestyle relevance (personal values), supporting manipulation of endorsed-value types. Posts highlighted the camera's "4K resolution" and communicated either personal (e.g., Taylor's vlogging experience with the camera), social (e.g., values attached by YouTubers and influencers), or original value claims (e.g., company-created product values; see Appendix B). A second pretest ($N = 261$) confirmed no significant differences in processing fluency ($p = 0.56$), message attitudes ($p = 0.27$), or product attitudes ($p = 0.52$) across value types.

Participants completed manipulation checks for influencer and endorsed-value types. Influencer type was assessed on a 7-point scale ("*Taylor is…*"; 1 = VIs, 7 = HIs). For VI conditions, perceived realism was measured using a three-item, 7-point scale (e.g., "*Taylor appeared cartoonish*"; Arsenyan & Mirowska, 2021; $\alpha = 0.79$). Endorser credibility was measured with four items (e.g., "*To what extent do you agree that Taylor is credible to endorse the camera?*"; Chaiken & Maheswaran, 1994; $\alpha = 0.89$), endorsed-value authenticity with four items (e.g., "*The values depicted in the post demonstrated that the influencer says what he truly thinks and believes about the product*"; Becker et al., 2019; $\alpha = 0.84$), and endorsed-value adoption with four items (e.g., "*The product values discussed in the post motivated me to incorporate these values into my life*"; $\alpha = 0.77$), all on 7-point scales (see Appendix E for full measures).

## 6.2. Results

### 6.2.1. Manipulation checks

All manipulations were successful. One-way ANOVA confirmed significant differences in perceived influencer type ($F(2, 903) = 675.90$, $p < 0.001$, $\eta^2 = 0.600$), with HI rated as more human than both VIs (all $ps < 0.001$), and high-realism VI rated as more realistic than low-realism VI ($p < 0.001$). ANOVAs also confirmed that each post was perceived as most representative of its intended value type (original, social, or personal; all $ps < 0.001$), confirming successful manipulation of endorsed-value type.

### 6.2.2. Hypothesis testing

*6.2.2.1. Direct effects (H1, H2, H3, H4). Endorsed-Value Adoption.* Influencer type did not significantly affect endorsed-value adoption ($p = 0.35$), contrary to H1. Endorsed-value type, however, significantly influenced adoption ($F(2, 903) = 4.76$, $p = 0.01$, $\eta^2 = 0.01$). Original values ($M = 4.01$, $SD = 1.13$) produced higher adoption than social ($M = 3.78$, $SD = 1.13$, $p = 0.04$) and personal values ($M = 3.75$, $SD = 1.04$, $p = 0.01$), with no difference between social and personal values ($p = 0.93$). These findings do not support H2, which predicts that personal values would drive the highest adoption.

*Endorser Credibility*. Supporting H3, influencer type significantly affected endorser credibility ($F(2, 903) = 23.74$, $p < 0.001$, $\eta^2 = 0.05$). High-realism VI ($M = 3.46$, $SD = 1.51$) was perceived as more credible than both HI ($M = 2.64$, $SD = 1.51$, $p < 0.001$) and low-realism VI ($M = 2.84$, $SD = 1.54$, $p < 0.001$), with no significant difference between HI and low-realism VI ($p = 0.25$). In line with H4, endorsed-value type had no significant direct effect on endorser credibility ($p = 0.08$).

Robustness checks confirmed no significant differences in demographics for the effects on either endorsed-value adoption or endorser credibility.

*6.2.2.2. Mediation analysis: The authenticity-credibility paradox (H5, H6, H7)*. Sequential mediation analysis (PROCESS Model 6) tested the competing indirect effects of credibility and authenticity (see Table 2). Results demonstrate that endorser credibility can offset the authenticity deficit inherent in VIs, but this compensatory mechanism is significant only for high-realism VI.

Specifically, the indirect effect via endorsed-value authenticity alone was significantly negative for both low-realism ($B = -0.075$) and high-realism VIs ($B = -0.140$). The stronger negative effect for high-realism VI supports H6a, confirming that they suffer a greater authenticity penalty. The indirect effect through endorser credibility alone was positive and significant for high-realism VI ($B = 0.116$), but not for low-realism VI, supporting H5a and highlighting the unique credibility advantage of high-realism VIs in driving value adoption. Finally, the sequential indirect effect (influencer type → credibility → authenticity → adoption) was strong and positive for high-realism VI ($B = 0.230$), but non-significant for low-realism VI, providing support for H7.

These findings support resolving the authenticity-credibility paradox. VIs create an authenticity deficit (supporting H6), but high-realism VI uniquely benefit from a credibility advantage (supporting H5) that is powerful enough to sequentially restore authenticity perceptions and ultimately drive endorsed-value adoption (supporting H7).

**Table 2**
Indirect effects of influencer type on endorsed-value adoption.

| Paths | Indirect Effect (SE) | Bootstrapping CI Lower Limit | Bootstrapping CI Upper Limit |
|---|---|---|---|
| Influencer Type → Endorser Credibility → Endorsed-Value Adoption | | | |
| • Low-realism VIs | 0.028 (0.019) | −0.007 | 0.065 |
| • High-realism VIs | 0.116 (0.027) | 0.071 | 0.170 |
| Influencer Type → Endorsed-Value Authenticity → Endorsed-Value Adoption | | | |
| • Low-realism VIs | −0.075 (0.035) | −0.146 | −0.007 |
| • High-realism VIs | −0.140 (0.035) | −0.210 | −0.073 |
| Influencer Type → Endorser Credibility → Endorsed-Value Authenticity → Endorsed-Value Adoption | | | |
| • Low-realism VIs | 0.056 (0.035) | −0.013 | 0.124 |
| • High-realism VIs | 0.230 (0.038) | 0.157 | 0.308 |

*6.2.2.3. Moderated mediation by endorsed-value type (H8, H8a).* To assess whether endorsed-value type moderates the effectiveness of credibility repairing authenticity, we employed PROCESS Model 91. Results revealed a significant moderation effect ($p = 0.046$), indicating that the strength of the credibility-to-authenticity pathway varies by the type of value endorsed, supporting H8. However, the moderation effect was not strongest for social values as predicted in H8a. For high-realism VI, the strongest mediation occurred for personal values ($B = 0.29$, $SE = 0.05$, 95 % CI = [.19, 0.38]), followed by social ($B = 0.24$, $SE = 0.04$, 95 % CI = [.16, 0.33]) and original values ($B = 0.24$, $SE = 0.04$, 95 % CI = [.16, 0.32]). The index of moderated mediation was only significant for personal values. For low-realism VIs, the moderated mediation effect was not significant across three value types.

### 6.3. Discussion

Study 1 demonstrates the authenticity-credibility paradox in VI marketing and clarifies how consumers resolve it. Our findings move beyond the human versus machine debate (Ma & Li, 2024) by showing that high-realism VIs are perceived as less authentic but more credible than HIs. Mediation analysis indicates that while high-realism VIs face an initial authenticity penalty, their perceived high credibility serves as a psychological repair, restoring authenticity perceptions and enhancing endorsed-value adoption. This dynamic positions credibility as an active psychological tool consumers use to address scepticism toward artificial endorsers, thereby extending source credibility theory (Lou & Yuan, 2019).

The findings reveal no consistent direct advantage for HIs in driving endorsed-value adoption, challenging assumptions of human superiority. High-realism VIs matched HIs in driving adoption, suggesting that their perceived competence and consistency can rival human relatability in digital contexts. This insight supports the CASA paradigm (Nass & Moon, 2000), suggesting that consumers increasingly accept virtual agents as credible social actors. Unexpectedly, company-originated values were more effective than personal or social values in promoting adoption, indicating that clear, functional claims may be more persuasive for VIs than abstract values. These results suggest that the effectiveness of personal values observed in HI marketing (Zhou et al., 2021) may not generalise to VIs, highlighting influencer type differences in endorsing value propositions to drive favorable consumer responses.

Building on these findings, our moderation analysis further clarifies the nuanced role of endorsed-value type in VI marketing. Specifically, the effectiveness of the repair mechanism depends on the value proposition of prompted products. High-realism VIs proved most effective at driving endorsed-value adoption when endorsing personal values, possibly because their idealised characteristics resonate with consumers' aspirational self-concepts, providing greater source validation to compensate for low authenticity.

While Study 1 establishes the existence and resolution of the paradox, it treats credibility and authenticity as broad constructs. The specific dimensions of credibility (e.g., expertise, trustworthiness) that most effectively repair authenticity deficits remain unclear. Addressing this limitation requires a more granular analysis, which we undertake in Study 2.

## 7. Study 2

Study 2 builds on Study 1 by examining how specific components of endorser credibility, expertise, trustworthiness, and attractiveness, interact with different dimensions of authenticity to shape consumer responses. It also assesses the moderating roles of endorsed-value type and perceived value in the authenticity repairing process and final persuasion.

### 7.1. Method

A total of 613 participants (57 % female; $M_{age} = 40.54$) were recruited via Prolific and randomly assigned to a 3 (influencer type: HI, high-realism VI, low-realism VI) × 3 (endorsed-value type: original, social, personal) between-subjects design. After excluding those who failed attention checks, 597 responses were included in the analysis. Each participant viewed the profile of "Mike," a fictional male influencer modelled after the real-life VI Phoenix McEwan, with the image for both HI and high-realism VI generated using the same AI tool as in Study 1 (see Appendix C). The low-realism VI used a stylised version of the same base image, with cartoon-like features and reduced photographic detail to visibly signal artificiality while retaining core identity cues. A pretest ($N = 151$) confirmed no significant differences in perceived familiarity ($p = 0.83$) or attitudes toward the profiles ($p = 0.51$) across influencer types.

Participants then viewed an Instagram post where Mike endorsed a dietary supplement, a category chosen for its relevance to influencer marketing and its capacity to elicit both utilitarian (original) and symbolic (social, personal) value perceptions. The product feature, supporting stress release, was held constant across conditions, while messaging was tailored to manipulate endorsed-value type (see Appendix D). For example, the social value condition focused on group benefits for office workers, whereas the personal value condition highlighted individual experiences. A second pretest ($N = 151$) confirmed no differences in processing fluency ($p = 0.91$), message attitude ($p = 0.59$), or product attitude ($p = 0.22$) across conditions.

Following exposure, participants completed manipulation checks for perceived influencer type (perceived realism: $\alpha = 0.76$) and endorsed-value type, using the same validated scales as in Study 1. They rated Mike on attractiveness ($\alpha = 0.91$), trustworthiness ($\alpha = 0.93$), and expertise ($\alpha = 0.89$), with overall credibility computed as their mean ($\alpha = 0.94$). Multiple dimensions of authenticity were assessed. Endorsed-value authenticity ($\alpha = 0.83$) was assessed by using the same scale as Study 1. Product authenticity was measured using a three-item scale ($\alpha = 0.86$). Message authenticity ($\alpha = 0.74$) was operationalised as the mean of perceived brand control ($\alpha = 0.77$), motivation of messages ($\alpha = 0.81$), and message accuracy ($\alpha = 0.74$). Influencer authenticity ($\alpha = 0.94$) was assessed as the mean of genuineness ($\alpha = 0.78$), identity disclosure ($\alpha = 0.64$), sincerity ($\alpha = 0.87$), cognitive experience ($\alpha = 0.84$), and conscious emotionality ($\alpha = 0.93$). Perceived value ($\alpha = 0.98$) was calculated as the mean of self-expression ($\alpha = 0.82$), social signalling ($\alpha = 0.73$), and emotional attachment ($\alpha = 0.75$). Purchase intention ($\alpha = 0.96$) and eWOM intention ($\alpha = 0.88$) were also measured. All items (see Appendix E) used 7-point Likert scales and were presented in randomised order to minimise response bias.

### 7.2. Results

#### 7.2.1. Manipulation checks

All manipulations were successful. ANOVA confirmed significant differences in perceived influencer type ($F(2, 594) = 103.68$, $p < 0.001$, $\eta^2 = 0.26$), with HI perceived as more human than both VIs (all $ps < 0.001$), and high-realism VI rated as more realistic than low-realism VI ($p < 0.001$). Posts were also rated as most representative of their assigned value type (original, social, or personal; all $ps < 0.001$), confirming the effectiveness of value manipulations.

#### 7.2.2. Hypothesis testing

*7.2.2.1. Direct effects (H1, H2, H3, H4). Purchase and eWOM Intentions.* Contrary to H1 predicting HIs' superiority, high-realism VI ($M = 2.23$, $SD = 1.48$) generated higher purchase intention than HI ($M = 1.90$, $SD = 1.35$, $p = 0.04$; $F(2, 594) = 3.22$, $p = 0.04$, $\eta^2 = 0.01$), with no significant differences between HI and low-realism VI ($M = 1.98$, $SD = 1.28$, $p =$

0.83) or between the two VIs ($p = 0.15$). Influencer type did not significantly affect eWOM intention, further rejecting H1. H2 was also unsupported, as endorsed-value type had no significant direct effect on purchase intention ($p = 0.35$) or eWOM ($p = 0.89$).

*Endorser Credibility*. Supporting H3, influencer type significantly impacted endorser credibility ($F(2, 594) = 23.86$, $p < 0.001$, $\eta^2 = 0.07$). High-realism VI ($M = 3.65$, $SD = 1.02$) was rated as more credible than both HI ($M = 2.90$, $SD = 1.13$, $p < 0.001$) and low-realism VI ($M = 3.15$, $SD = 1.11$, $p < 0.001$), with no significant differences between HI and low-realism VI ($p = 0.06$). Consistent with H4, endorsed-value type did not affect endorser credibility ($p = 0.11$).

ANCOVA involving demographic covariates produced non-significant changes, confirming the robustness of these findings for purchase intention, eWOM intention, and endorser credibility.

*7.2.2.2. Mediation analysis: deconstructing authenticity-credibility paradox (H5, H6, H7)*. To deconstruct the authenticity-credibility paradox, we conducted a comprehensive serial mediation analysis (PROCESS Model 6), with results detailed in Table 3. Consistent with H6, both high-realism and low-realism VIs, compared to HI, produced stronger negative indirect effects on purchase and eWOM intentions through endorsed-value authenticity ($B_{purchase} = -0.17$, $B_{eWOM} = -0.09$), product authenticity ($B_{purchase} = -0.18$, $B_{eWOM} = -0.10$), and message authenticity ($B_{purchase} = -0.06$, $B_{eWOM} = -0.04$). This authenticity penalty was greatest for high-realism VI, supporting H6a. Interestingly, high-realism VI positively impacted purchase and eWOM intentions through enhanced influencer authenticity ($B_{purchase} = 0.07$, $B_{eWOM} = 0.06$), though this effect was absent for low-realism VI. At the same time, consistent with H5 and H5a, high-realism VI generated stronger positive indirect effects on purchase and eWOM intentions through endorser credibility than either low-realism VI or HI. These findings establish the paradox that VIs benefit from a credibility advantage yet suffer from authenticity deficits.

The full sequential mediation pathway, where credibility restores authenticity and then drives behavioural intentions, was significantly positive across all four authenticity dimensions for both VIs (see Table 3). Notably, the positive effects were consistently and substantially stronger for high-realism VI. For example, the indirect effect via credibility and product authenticity on purchase intention was nearly three times greater for high-realism VI ($B = 0.251$) than for low-realism VI ($B = 0.085$). These findings support H7 and demonstrate that credibility effectively offsets authenticity deficit to drive positive consumer responses.

To further clarify the mechanism underlying the dissonance-reduction effect predicted in H7, we decomposed endorser credibility into expertise, trustworthiness, and attractiveness using PROCESS Model 80. The results in Table 4 revealed that the three credibility dimensions play distinct roles.

Trustworthiness and expertise drive the sequential mediation effect for high-realism VI, resulting in significant positive impacts on purchase and eWOM intentions across claim-based authenticity (endorsed-value, product, and message authenticity), except only eWOM intention for influencer authenticity. These pathways were largely non-significant for low-realism VI.

Attractiveness is a double-edged sword. While it produced a small positive effect via endorsed-value authenticity, it also led to negative indirect effects through message authenticity for both VI types (e.g., for high-realism VI on purchase intention: $B = -0.017$; for low-realism VI on purchase intention: $B = -0.013$). These findings suggest that while aesthetically pleasing, high attractiveness can diminish perceived message sincerity.

*7.2.2.3. Moderated mediation by endorsed-value type (H8, H8a)*. We examined whether endorsed-value type moderates credibility's ability to repair authenticity using PROCESS Model 91. Results show a significant moderation effect for product authenticity ($p_{purchase\ \&\ eWOM} = 0.02$) and a marginally significant effect for endorsed-value authenticity ($p_{purchase\ \&\ eWOM} = 0.05$). These findings indicate that credibility's impact on authenticity depends on the value being endorsed, supporting H8. Social values consistently generated stronger conditional indirect effects than original or personal values for both VI types. For example, repairing product authenticity for low-realism VIs on purchase intention was highest for social values ($B = 0.31$) compared to original ($B = 0.22$) and personal values ($B = 0.23$). This pattern held across all outcomes, with the index of moderated mediation significant for social value in repairing product and endorsed-value authenticity. Therefore, the directional prediction of H8a is supported, with social values yielding the strongest effects.

*7.2.2.4. Moderated mediation by perceived values (H9, H9a)*. Consistent with H9 and H9a, moderated mediation analyses (PROCESS Model 87) showed that the effect of credibility-restored authenticity on behavioural intention strengthened as participants' perceived value of the product increased. This moderating effect was observed for endorsed-value ($B_{purchase} = 0.21$, $p < 0.001$; $B_{eWOM} = 0.18$, $p < 0.001$) and product authenticity ($B_{purchase} = 0.18$, $p < 0.001$; $B_{eWOM} = 0.15$, $p < 0.001$) on purchase and eWOM intentions. For both high and low-realism VIs, these effects were mostly significant at moderate to high

**Table 3**
Serial mediation effect of endorser credibility and authenticity dimensions on purchase and eWOM intentions.

| Paths | Purchase Intention | Bootstrapping CI | | eWOM Intention | Bootstrapping CI | |
|---|---|---|---|---|---|---|
| | **Indirect Effect** (SE) | Lower Limit | Upper Limit | **Indirect Effect** (SE) | Lower Limit | Upper Limit |
| Influencer Type → Endorser Credibility → **Endorsed-Value Authenticity** → Purchase/eWOM intentions | | | | | | |
| • Low-realism VIs | 0.085 (0.040) | 0.001 | 0.170 | 0.046 (0.022) | 0.005 | 0.095 |
| • High-realism VIs | 0.253 (0.051) | 0.164 | 0.361 | 0.135 (0.034) | 0.072 | 0.207 |
| Influencer Type → Endorser Credibility → **Product Authenticity** → Purchase/eWOM intentions | | | | | | |
| • Low-realism VIs | 0.085 (0.039) | 0.010 | 0.165 | 0.049 (0.023) | 0.006 | 0.096 |
| • High-realism VIs | 0.251 (0.045) | 0.168 | 0.343 | 0.144 (0.030) | 0.089 | 0.207 |
| Influencer Type → Endorser Credibility → **Influencer Authenticity** → Purchase/eWOM intentions | | | | | | |
| • Low-realism VIs | 0.046 (0.027) | 0.003 | 0.107 | 0.042 (0.024) | 0.003 | 0.097 |
| • High-realism VIs | 0.135 (0.051) | 0.041 | 0.238 | 0.125 (0.045) | 0.042 | 0.219 |
| Influencer Type → Endorser Credibility → **Message Authenticity** → Purchase/eWOM intentions | | | | | | |
| • Low-realism VIs | 0.047 (0.024) | 0.005 | 0.098 | 0.032 (0.017) | 0.004 | 0.069 |
| • High-realism VIs | 0.140 (0.035) | 0.078 | 0.214 | 0.096 (0.029) | 0.044 | 0.156 |

**Table 4**
Serial mediation effect of endorser credibility and authenticity dimensions on purchase and eWOM intentions.

| Paths | Purchase Intention | | | eWOM Intention | | |
|---|---|---|---|---|---|---|
| | | Bootstrapping CI | | | Bootstrapping CI | |
| | **Indirect Effect** (SE) | Lower Limit | Upper Limit | **Indirect Effect** (SE) | Lower Limit | Upper Limit |
| **Endorsed-Value Authenticity** | | | | | | |
| Influencer Type → **Expertise** → Endorsed-Value Authenticity → Purchase/eWOM intentions | | | | | | |
| • Low-realism VIs | 0.014 (0.013) | −0.011 | 0.040 | 0.007 (0.007) | −0.006 | 0.022 |
| • High-realism VIs | 0.079 (0.023) | 0.039 | 0.131 | 0.041 (0.015) | 0.016 | 0.073 |
| Influencer Type → **Trustworthiness** → Endorsed-Value Authenticity → Purchase/eWOM intentions | | | | | | |
| • Low-realism VIs | 0.020 (0.024) | −0.028 | 0.068 | 0.010 (0.013) | −0.014 | 0.038 |
| • High-realism VIs | 0.130 (0.034) | 0.073 | 0.204 | 0.068 (0.022) | 0.030 | 0.115 |
| Influencer Type → **Attractiveness** → Endorsed-Value Authenticity → Purchase/eWOM intentions | | | | | | |
| • Low-realism VIs | 0.019 (0.009) | 0.004 | 0.040 | 0.010 (0.005) | 0.002 | 0.022 |
| • High-realism VIs | 0.025 (0.011) | 0.006 | 0.050 | 0.013 (0.007) | 0.003 | 0.028 |
| **Product Authenticity** | | | | | | |
| Influencer Type → **Expertise** → Product Authenticity → Purchase/eWOM intentions | | | | | | |
| • Low-realism VIs | 0.018 (0.018) | −0.016 | 0.056 | 0.010 (0.010) | −0.008 | 0.032 |
| • High-realism VIs | 0.106 (0.030) | 0.053 | 0.171 | 0.060 (0.019) | 0.028 | 0.100 |
| Influencer Type → **Trustworthiness** → Product Authenticity → Purchase/eWOM intentions | | | | | | |
| • Low-realism VIs | 0.019 (0.024) | −0.029 | 0.067 | 0.011 (0.014) | −0.015 | 0.038 |
| • High-realism VIs | 0.126 (0.033) | 0.069 | 0.195 | 0.070 (0.020) | 0.036 | 0.114 |
| Influencer Type → **Attractiveness** → Product Authenticity → Purchase/eWOM intentions | | | | | | |
| • Low-realism VIs | 0.002 (0.010) | −0.018 | 0.023 | 0.001 (0.006) | −0.010 | 0.013 |
| • High-realism VIs | 0.003 (0.014) | −0.024 | 0.030 | 0.001 (0.008) | −0.013 | 0.017 |
| **Influencer Authenticity** | | | | | | |
| Influencer Type → **Expertise** → Influencer Authenticity → Purchase/eWOM intentions | | | | | | |
| • Low-realism VIs | 0.008 (0.009) | −0.007 | 0.029 | 0.007 (0.008) | −0.007 | 0.026 |
| • High-realism VIs | 0.044 (0.025) | −0.003 | 0.095 | 0.043 (0.022) | 0.002 | 0.089 |
| Influencer Type → **Trustworthiness** → Influencer Authenticity → Purchase/eWOM intentions | | | | | | |
| • Low-realism VIs | 0.006 (0.010) | −0.010 | 0.030 | 0.006 (0.009) | −0.011 | 0.027 |
| • High-realism VIs | 0.043 (0.025) | −0.003 | 0.095 | 0.042 (0.023) | 0.002 | 0.092 |
| Influencer Type → **Attractiveness** → Influencer Authenticity → Purchase/eWOM intentions | | | | | | |
| • Low-realism VIs | 0.004 (0.003) | −0.001 | 0.013 | 0.004 (0.003) | −0.002 | 0.012 |
| • High-realism VIs | 0.006 (0.004) | −0.001 | 0.016 | 0.006 (0.004) | −0.003 | 0.016 |
| **Message Authenticity** | | | | | | |
| Influencer Type → **Expertise** → Message Authenticity → Purchase/eWOM intentions | | | | | | |
| • Low-realism VIs | 0.009 (0.009) | −0.007 | 0.030 | 0.006 (0.006) | −0.005 | 0.020 |
| • High-realism VIs | 0.053 (0.020) | 0.020 | 0.098 | 0.035 (0.015) | 0.010 | 0.071 |
| Influencer Type → **Trustworthiness** → Message Authenticity → Purchase/eWOM intentions | | | | | | |
| • Low-realism VIs | 0.013 (0.016) | −0.019 | 0.046 | 0.008 (0.011) | −0.013 | 0.032 |
| • High-realism VIs | 0.084 (0.026) | 0.041 | 0.140 | 0.056 (0.020) | 0.021 | 0.100 |
| Influencer Type → **Attractiveness** → Message Authenticity → Purchase/eWOM intentions | | | | | | |
| • Low-realism VIs | −0.013 (0.007) | −0.028 | −0.003 | −0.009 (0.005) | −0.020 | −0.001 |
| • High-realism VIs | −0.017 (0.008) | −0.035 | −0.004 | −0.011 (0.006) | −0.025 | −0.002 |

levels. These findings confirm that authenticity repair process is most effective in driving behavioural intentions when consumers find the endorsed value as personally meaningful.

### 7.3. Discussion

Study 2 clarifies the psychological mechanism underlying the authenticity-credibility paradox in VIs. High-realism VIs incur a marked authenticity penalty on claim-based dimensions such as endorsed-value, product, and message authenticity, reflecting consumer scepticism about their artificiality. Yet, these VIs receive the highest ratings for influencer authenticity, indicating that consumers recognise consistency and coherence in VIs' digital personas even as they remain wary of commercial claims. This decoupling advances VI literature (Ma & Li, 2024) by suggesting that consumers distinguish between the messenger and the message, a key insight for understanding how authenticity operates for artificial agents.

The analysis of credibility dimensions reveals distinct strategic levers for managing this paradox in high-realism VIs. Trustworthiness and expertise emerge as the primary mechanisms for repairing deficits in claim-based authenticity and reinforcing influencer authenticity, directly addressing consumer scepticism. These findings suggest that positioning VIs as knowledgeable and trustworthy enhances the coherence of their digital personas. Attractiveness, however, presents a trade-off. While attractiveness addresses endorsed-value authenticity, it increased message authenticity penalty, likely by triggering perceptions of manipulative intent as described by the persuasion knowledge model (Friestad & Wright, 1994). This insight highlights the importance of

balancing aesthetic design with sincerity in VI strategy.

Consistent with Study 1, we find no direct advantage for HI, and personal value does not outperform other value types in driving behavioural intentions or credibility. Instead, the principal effect of endorsed value type is to moderate the capability of credibility in addressing authenticity deficits, as evidenced in our moderated mediation analyses. The findings highlight the strategic importance of credibility management in VI campaigns promoting social values and aiming to enhance purchase and eWOM.

Perceived value also operates as a boundary condition. Credibility-restored authenticity significantly drives behavioural outcomes only when consumers perceive the product as valuable. This finding indicates a critical marketing insight that even optimal authenticity management cannot compensate for low product value. Overall, Study 2 demonstrates that while VI authenticity deficits can be offset by trustworthiness and expertise, these strategies are effective only when endorsements are tied to products that consumers genuinely value.

## 8. General discussion

### 8.1. Theoretical contributions

This research primarily extends cognitive dissonance theory (Festinger, 1957) beyond its traditional applications in attitude-behaviour consistency and post-purchase rationalisation, applying it to persuasion by non-human agents in VI marketing. We show that credibility acts as a dissonance-reduction mechanism, enabling consumers to reconcile high credibility with low authenticity and maintain psychological consistency. By introducing the sequential credibility-to-authenticity repair pathway, we reveal a novel psychological process that has not been previously identified in the cognitive dissonance literature. This research advances understanding across domains of VI marketing, authenticity, and persuasion.

#### 8.1.1. Resolving the authenticity-credibility paradox

We resolve the authenticity-credibility paradox in VI marketing (Lou et al., 2023; Sands et al., 2022). Unlike HIs, VIs are perceived as highly credible yet fundamentally inauthentic, reversing the traditional authenticity-to-credibility relationship established for HIs (Wang & Weng, 2024). Our results show that, for VIs, credibility functions as a psychological tool to repair authenticity deficits. Extending cognitive dissonance theory (Festinger, 1957), we demonstrate that consumers use credibility cues to offset scepticism about VIs' artificiality, restoring authenticity perceptions and enabling persuasion.

Our results also clarify that distinct facets of credibility address different authenticity deficits. Trustworthiness and expertise repair claim-based authenticity shortfalls, such as endorsed-value, product, and message authenticity, as well as strenthening influencer authenticity, by lending coherence and authority to virtual personas and reducing consumer scepticism (Munnukka et al., 2022; Dondapati & Dehury, 2024). Attractiveness, in contrast, presents a compromise. It can repair endorsed-value authenticity deficiency but may reduce message authenticity by activating persuasion knowledge and prompting suspicion about intent (Friestad & Wright, 1994). This differentiation highlights specific psychological levers for managing the authenticity-credibility trade-off in VI campaigns.

#### 8.1.2. Advancing authenticity theory for non-human agents

This research advances authenticity theory by demonstrating its multidimensional nature in VI contexts. While prior research often treats authenticity as a unified construct (e.g., Cheah et al., 2024; Kim & Wang, 2024), our findings show that different dimensions can diverge. High-realism VIs receive the highest ratings for influencer authenticity, reflecting perceived transparency, genuineness, and consistency in persona, but the lowest for claim-based authenticity, such as endorsed value, product, and message authenticity that indicate the truthfulness of endorsements. This divergence stems from consumers evaluating the messenger and the message separately, reframing authenticity as a dynamic, potentially conflicted process. This perspective provides a novel theoretical foundation for managing diverse authenticity of artificial agents.

#### 8.1.3. Defining the role of anthropomorphism in shaping credibility and authenticity

Our work extends anthropomorphism theory by identifying both credibility rewards and authenticity penalty linked to VI realism. High-realism VIs elevate consumers' expectations for human-like qualities (Thomas & Fowler, 2021; Sands et al., 2022). When these expectations are reduced by detectable artificiality, consumers penalise claim-based authenticity. Conversely, low-realism VIs set more modest expectations and thus experience less of an authenticity penalty. These findings clarify how increased realism can enhance credibility but also paradoxically damage authenticity, extending current perspectives on anthropomorphism in VI marketing by demonstrating its dual and often conflicting effects on consumer perception.

#### 8.1.4. Theorising the evolving dynamics of digital influence

Our findings challenge the digital influence literature by showing that high-realism VIs can now achieve higher credibility than HIs when their expertise and trustworthiness are emphasised, reversing prevailing emphasis on the superiority of HIs (e.g., Lou et al., 2023; Lee et al., 2024). This shift reflects ongoing advances in technology and evolving consumer perceptions, as the uncanny effect lessens and credibility cues become more salient. Consumers increasingly accept VIs as legitimate endorsers, suggesting that VIs' effectiveness is not a fixed comparison with humans but a dynamic process shaped by technological and cultural adaptation. These insights call for digital influence theories to account for the continual evolution of both technology and consumer attitudes.

#### 8.1.5. Delineating boundary conditions of the repair mechanism

We also advance the understanding of value proposition in VI marketing by clarifying the conditions under which the credibility-to-authenticity repair mechanism operates. Endorsed-value typecan alter the effectiveness of credibility in restoring product and endorsed-value authenticity. Social value endorsements enhance credibility's repair capability for driving purchase and eWOM intentions, while personal value endorsements are more effective for driving endorsed-value adoption. Additionally, consumer perceived value sets a boundary. The repaired product and endorsed-value authenticity only translate into behavioural intentions when consumers perceive the product as valuable (Liao & Chen, 2024). This result consistent with motivated reasoning theory (Kunda, 1990), suggesting that consumers are motivated to resolve authenticity concerns primarily when they perceive the product as enabling self-expression, social signalling, or emotional attachment. When perceived value is low, the psychological repair mechanism is not activated.

#### 8.1.6. Integrating theoretical contributions

Collectively, these contributions provide a comprehensive framework for managing artificial endorsers in digital marketing. By clarifying the authenticity-credibility paradox and unpacking the multidimensional nature of authenticity, we highlight where brands should target authenticity repair (product, message, endorsed-value domains) and where reinforcement is most effective (influencer persona domains). Our theorisation of the credibility rewards and authenticity penalties associated with VI realism, alongside the evolving dynamics of digital influence, provides actionable insights for brand management. This integrated perspective informs campaign design, influencer selection, and the optimal calibration of realism across various value propositions and product categories.

### 8.2. Managerial implications

This research provides actionable guidance for brands seeking to deploy VIs while managing the authenticity-credibility paradox.

#### 8.2.1. In what dimensions should authenticity be managed in VI marketing?

Marketers should treat authenticity as multidimensional, distinguishing influencer authenticity from claim-based authenticity, which includes message, product, and endorsed-value authenticity. High-realism VI excel at influencer authenticity, outperforming both low-realism VIs and HIs. To capitalise on this strength, brands should develop lifelike personas and narratives that emphasise genuineness, consistency, sincerity, and emotional resonance.

Although message, product, and endorsed-value authenticity are generally lower for VIs, these dimensions remain important for consumer response. To manage message and product authenticity, endorsed content should reduce explicit brand control and commercial-driven motivation, while prioritising accuracy and genuine product benefits. Endorsed-value authenticity depends on how well the values of the VI, brand, and product align. Marketers should consistently communicate authentic values and benefits that are coherent across all campaign elements.

#### 8.2.2. How to mitigate negative authenticity perceptions to foster optimal outcomes?

To address authenticity deficits, marketers should emphasise VI credibility by prioritising trustworthiness and expertise over attractiveness, which is often overemphasised in the industry. High-realism VIs are most effective when positioned as trustworthy and knowledgeable, addressing deficits across endorsed-value, product, and message authenticity, while also reinforcing influencer authenticity. Campaigns incorporating social values can further enhance this authenticity repair process, improving purchase and eWOM intentions, while personal value are more effective to drive endorsed-value adoption. To ensure favourable consumer response, campaigns should target consumers who find the product personally meaningful.

Moreover, campaign strategies should match the VI's realism. High-realism VIs are better suited to campaigns that require authority and expertise, such as those for luxury or technical products. Low-realism VIs are more effective in creative or playful campaigns, such as lifestyle or entertainment products, where authenticity expectations are lower, and flexibility is valued. Aligning the VI's realism, value proposition, and campaign objectives is essential to minimise authenticity gaps and maximise campaign impact.

### 8.3. Limitations and avenues for further research

While this research addresses a foundational paradox in VI marketing, several limitations and promising directions merit future exploration. First, while we identify key boundary conditions for the authenticity-credibility mechanism, such as endorsed-value type and perceived value, future research should examine additional moderators. These factors might include product involvement, consumer traits such as scepticism and need for cognition, and cultural differences that could shape the relative importance of credibility and authenticity.

Second, our high versus low realism design raises the question of curvilinear effects. VIs with moderate realism may achieve an optimal balance between human likeness and stylised aesthetics, minimising authenticity penalties while preserving credibility. Research exploring non-linear effects would deepen understanding of effective anthropomorphic design.

A further limitation is our use of male VIs as stimuli. While this choice reduced potential confounds related to sexualisation often associated with female influencers and ensured condition consistency, gender stereotypes may affect perceptions of expertise and trustworthiness. Future studies should test whether findings generalise across a broader range of demographic avatars, including female and child-like VIs.

We focus on immediate behavioural intentions following single exposures. Longitudinal research could assess how repeated interactions with VIs shape sustained eWOM and long-term engagement, and whether the dissonance-reduction process endures, strengthens, or fades over time.

Additionally, our findings are based on static evaluations. Future research should investigate VI effectiveness in dynamic, interactive contexts such as augmented reality, virtual reality, and AI-driven chatbots. These environments may introduce new dimensions of authenticity, such as interactive authenticity (the perceived genuineness of AI-driven engagement) or technological authenticity (the seamlessness of the digital rendering), whose relevance may depend on consumer digital literacy and context.

Our examination controls for brand identity and prior eWOM by using fictitious brands, improving internal validity, but may limit generalisability to establish brands with reputational histories. The credibility repair mechanism may be most effective for brands with new or neutral reputation, where consumers rely more on immediate influencer cues, and for brands with moderate positive equity, where VI credibility can reinforce favourable sentiment. In contrast, this mechanism may be less effective for brands with strong negative eWOM, where scepticism persists, or for iconic brands with extremely high equity, where brand reputation overshadows influencer effects. Future research should consider how brand history and existing consumer sentiment moderate the impact of influencer credibility and authenticity.

Finally, while our research centres on a cognitive pathway where credibility repairs authenticity deficits, future studies should examine whether affective pathways rooted in emotional connection and social pathways based on social proof and community validation, can effectively address authenticity deficits, thereby building a more comprehensive theory of managing authenticity of non-human agents in digital marketing.

## CRediT authorship contribution statement

**Amy Zhou:** Writing – review & editing, Writing – original draft, Validation, Supervision, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Yang Liu:** Writing – original draft, Methodology, Formal analysis, Data curation. **Yonglin Dai:** Writing – original draft, Methodology, Formal analysis, Data curation.

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## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Appendices.

### *Appendix A. Study 1: Influencer profiles*

| Human Influencer | High-Realism Virtual Influencer | Low-Realism Virtual Influencer |
|---|---|---|
| Taylor is a 21-year-old Japanese-American influencer based in Tokyo, Japan. He loves music and always dreams of standing on the big stage and presenting his music. His favourite music styles are pop, jazz, R&B, rock, and many more. These famous musicians within these genres inspire him daily to create new music and push him to keep pursuing his dreams. Especially since 2020, Taylor has taken time to work on his career as a musician. He is also fond of street fashion and works part-time as a model. A third big hobby is basketball. Luckily, he was invited as a VIP guest of a famous basketball team in the NBA bubble last year. | Taylor is a 21-year-old Japanese-American virtual influencer based in Tokyo, Japan. He loves music and always dreams of standing on the big stage and presenting his music. His favourite music styles are pop, jazz, R&B, rock, and many more. These famous musicians within these genres inspire him daily to create new music and push him to keep pursuing his dreams. Especially since 2020, Taylor has taken time to work on his career as a musician. He is also fond of street fashion and works part-time as a model. A third big hobby is basketball. Luckily, he was invited as a VIP guest of a famous basketball team in the NBA bubble last year. | Taylor is a 21-year-old Japanese-American virtual influencer based in Tokyo, Japan. He loves music and always dreams of standing on the big stage and presenting his music. His favourite music styles are pop, jazz, R&B, rock, and many more. These famous musicians within these genres inspire him daily to create new music and push him to keep pursuing his dreams. Especially since 2020, Taylor has taken time to work on his career as a musician. He is also fond of street fashion and works part-time as a model. A third big hobby is basketball. Luckily, he was invited as a VIP guest of a famous basketball team in the NBA bubble last year. |
|  |  |  |

### *Appendix B. Study 1: Summary of influencer posts*

| Human/High-Realism Virtual Influencers | | |
|---|---|---|
| **Original Value Post** | **Social Value Post** | **Personal Value Post** |

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(*continued*)

**Human/High-Realism Virtual Influencers**

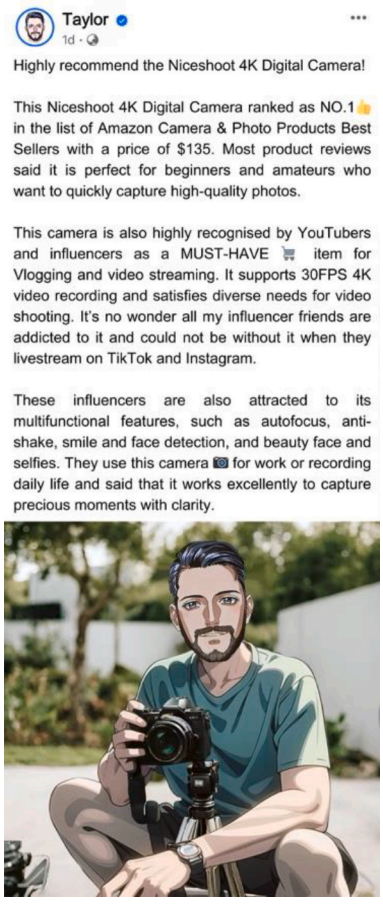
| Original Value Post | Social Value Post | Personal Value Post |
|---|---|---|
| Taylor<br>1d ·<br>Highly recommend the Niceshoot 4K Digital Camera!<br><br>The Niceshoot company launched its new 4K Digital Camera this October, which provides impressive image quality with 48 million pixels. The company sets its price at $135 and aims to make it easy to capture subjects and PERFECT for beginners and amateurs who want to quickly capture great moments and high-definition photos.<br><br>The company claims this camera is also excellent for Vlogging as it supports 30FPS 4K video recording. It can satisfy diverse needs for video shooting and provides excellent quality video streaming on social media such as Facebook, YouTube, TikTok, and Instagram.<br><br>This company also designed this smart camera with autofocus, anti-shake, smile and face detection, and beauty face and selfie. It effortlessly captures wonderful moments in daily life and travel and delivers perfect shooting results.<br>  | Taylor<br>1d ·<br>Highly recommend the Niceshoot 4K Digital Camera!<br><br>This Niceshoot 4K Digital Camera ranked as NO.1 in the list of Amazon Camera & Photo Products Best Sellers with a price of $135. Most product reviews said it is perfect for beginners and amateurs who want to quickly capture high-quality photos.<br><br>This camera is also highly recognised by YouTubers and influencers as a MUST-HAVE item for Vlogging and video streaming. It supports 30FPS 4K video recording and satisfies diverse needs for video shooting. It's no wonder all my influencer friends are addicted to it and could not be without it when they livestream on TikTok and Instagram.<br><br>These influencers are also attracted to its multifunctional features, such as autofocus, anti-shake, smile and face detection, and beauty face and selfies. They use this camera for work or recording daily life and said that it works excellently to capture precious moments with clarity.<br>  | Taylor<br>1d ·<br>Highly recommend the Niceshoot 4K Digital Camera!<br><br>As someone who takes many pictures and enjoys taking videos, I brought a range of cameras for photography and video and returned several of them. I have to say that the Niceshoot 4K Digital Camera , which is priced at $135, has more than lived up to my expectations. It is SUPERISING how clear and precise the images are, regardless of whether I am taking pictures of landscapes, portraits, or close-ups.<br><br>I also love to use this camera for Vlogging. It produces videos that are exceptionally crisp and detailed and can instantly latch onto targets thanks to the autofocus, which enables me to quickly catch ephemeral moments. This function is very helpful when I'm moving or in dynamic scenarios. I'm never going to miss the opportunity to shoot the ideal video.<br><br>Now I'm bringing it to my brother's wedding. I feel more confident with using it to capture meaningful and joyful moments for my family and friends!<br>  |

**Low-Realism Virtual Influencer**

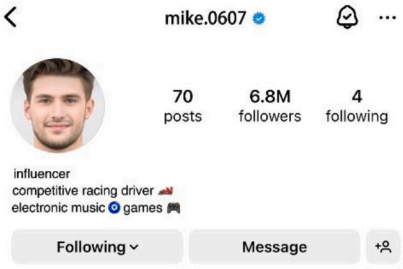
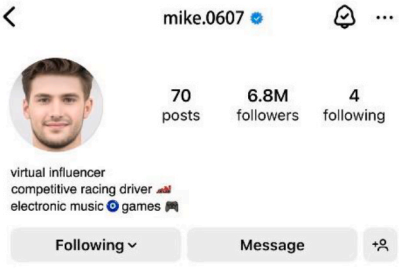
| Original Value Post | Social Value Post | Personal Value Post |
|---|---|---|

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(*continued*)

| Human/High-Realism Virtual Influencers<br>Original Value Post | Social Value Post | Personal Value Post |
|---|---|---|
|  |  |  |

*Appendix C. Study 2: Influencer profiles*

| Human Influencer | High-Realism Virtual Influencer | Low-Realism Virtual Influencer |
|---|---|---|
| Mike is a human influencer who burst onto the scene in 2019. Hailing from the vibrant city of London, Mike embodies the spirit of a competitive racing driver. With an unwavering determination, his ultimate aspirations are to immerse himself in prestigious race series, be the world champion in the racing world, and secure esteemed manufacturer endorsements. Mike's interests extend far beyond the racetrack. He spends his time listening to electronic music and gaming. He lives his life by three basic rules: 1. don't judge on appearances. 2. be open-minded. 3. always have a positive outlook. | Mike is a virtual influencer who burst onto the scene in 2019. Hailing from the vibrant city of London, Mike embodies the spirit of a competitive racing driver in both the virtual and real world. With an unwavering determination, his ultimate aspirations are to immerse himself in prestigious race series, be the world champion in the racing world, and secure esteemed manufacturer endorsements. Mike's interests extend far beyond the racetrack. He spends his time listening to electronic music and gaming. He lives his life by three basic rules: 1. don't judge on appearances. 2. be open-minded. 3. always have a positive outlook. | Mike is a virtual influencer who burst onto the scene in 2019. Hailing from the vibrant city of London, Mike embodies the spirit of a competitive racing driver in both the virtual and real world. With an unwavering determination, his ultimate aspirations are to immerse himself in prestigious race series, be the world champion in the racing world, and secure esteemed manufacturer endorsements. Mike's interests extend far beyond the racetrack. He spends his time listening to electronic music and gaming. He lives his life by three basic rules: 1. don't judge on appearances. 2. be open-minded. 3. always have a positive outlook. |
|  |  |  |

*Appendix D. Study 2: Summary of influencer posts*

**Human/High-Realism Virtual Influencers**

**Original Value Post**




**Social Value Post**




**Personal Value Post**




**Low-Realism Virtual Influencer**

**Original Value Post**




**Social Value Post**




**Personal Value Post**




*Appendix E. Summary of variable measurements*

| Variables | Variable Dimensions | Items | Reference |
|---|---|---|---|
| **Influencer Type** | **Influencer Type** | • Taylor is: (1 = a virtual influencer, 7 = a human influencer). | / |
| | **Realism** | • Taylor appeared genuinely human.<br>• Taylor appeared cartoonish.*<br>• Taylor's appearance is exaggerated.* | (Arsenyan & Mirowska, 2021) |
| **Endorsed-Value Type** | / | • After reading the post, to what extent do you think it introduces the original values?<br>• After reading the post, to what extent do you think it introduces the social values?<br>• After reading the post, to what extent do you think it introduces the personal values? | / |
| **Endorser Credibility** (Study 1) | / | • To what extent do you agree that Taylor is credible to endorse the camera?<br>• To what extent do you agree that Taylor is reliable to endorse the camera?<br>• To what extent do you agree that Taylor is trustworthy to endorse the camera?<br>• To what extent do you agree that Taylor is an expert in endorsing the camera? | (Chaiken & Maheswaran, 1994) |
| **Endorsed-Value Authenticity** | / | • The product values introduced in the post were authentic.<br>• The product values introduced in the post were truthful.<br>• The product values introduced in the post seemed genuine.<br>• The product values introduced in the post demonstrated that the influencer says what he truly thinks and believes about the product. | (Becker et al., 2019) |
| **Endorsed-Value Adoption** | / | • The product values discussed in the post enhanced my knowledge of the camera.<br>• The product values discussed in the post made it easier for me to recognise the value of the camera.<br>• The product values discussed in the post enhanced my ability to make a decision.<br>• The product values discussed in the post motivated me toincorporate these values into my life. | (Fang, 2014) |
| **Purchase Intention** | / | • How likely would you consider purchasing the camera mentioned in the post?<br>• How definitely would you consider purchasing the camera mentioned in the post?<br>• How probable would you consider purchasing the camera mentioned in the post? | (De Jans et al., 2018) |
| **eWOM Intention** | / | • I am interested in sharing this post with my friends/family on social media platforms.<br>• I will recommend this post to others.<br>• I am interested in liking and commenting on this post.<br>• I am willing to spread information, comments and experience about this product on social media.<br>• I am willing to share this post on my social media pages. | (Boerman et al., 2017) |
| **Endorser Credibility** (Study 2) | **Attractiveness** | • I think Mike is quite pretty.<br>• I find Mike very attractive physically.<br>• Mike looks stylish.<br>• Mike's appearance is quite attractive. | (Kim & Park, 2023) |
| | **Trustworthiness** | • I feel Mike is honest.<br>• I consider Mike trustworthy.<br>• I feel Mike is reliable.<br>• I feel Mike is dependable.<br>• I feel Mike is sincere. | (Ohanian, 1990) |
| | **Expertise** | • I feel Mike is knowledgeable.<br>• I feel Mike is qualified.<br>• I consider Mike an expert in his area.<br>• I think Mike is skilled.<br>• I think Mike is experienced. | (Ohanian, 1990) |
| **Product Authenticity** | / | • I think this product can perform the described functions.<br>• I think this product works in the same way as described in the post.<br>• I think the probability that this product matches the description in the post is high. | (Kalaignanam et al., 2013) |
| **Message Authenticity** | **Brand Control** | • This post created by Mike is supervised by the company of the posted product.*<br>• The information Mike wishes to share with followers is filtered by the company of the posted product.* | (Martínez-López et al., 2020) |
| | **Motivation of Messages** | • Mike recommends the product because he cares about his followers.<br>• Mike is genuinely concerned for the welfare of his followers.<br>• Mike recommends the product because he is passionate about it.<br>• Mike recommends the product for getting paid.*<br>• Mike recommends the product because he only cares about his profits.*<br>• Mike recommends the product to create a positive self-image.* | (Shan et al., 2020) |
| | **Message Accuracy** | • This post was inaccurate.*<br>• This post was exaggerated.*<br>• This post was overstated.* | (Becker et al., 2019) |
| **Influencer Authenticity** | **Genuineness** | • In the post, Mike describes actual experiences and interactions with the product.<br>• In the post, Mike gives his genuine opinions about the product.<br>• In the post, Mike shows his true beliefs about the product. | (Lee & Eastin, 2021) |
| | **Identity Disclosure** | • Mike presents a genuine and authentic version of himself.<br>• Mike shares behind-the-scenes content to give a glimpse into his life.<br>• Mike transparently discloses who he really is on social media. | (Koles et al., 2024) |
| | **Sincerity** | • Mike seems kind and good hearted.<br>• Mike is sincere.<br>• Mike comes off as very genuine.<br>• Mike is down-to-earth. | (Lee & Eastin, 2021) |
| | **Cognitive Experience** | • Mike can engage in a great deal of thought.<br>• Mike can imagine things on his own.<br>• Mike is capable of reasoning | (Dabiran et al., 2024) |
| | **Conscious Emotionality** | • Mike can experience remorse over actions which it deems to be shameful.<br>• Mike can experience compassion for people who feel down.<br>• Mike can experience guilt when he hurts someone with his behaviour.<br>• Mike can experience shame when people have negative views and judgments about him. | (Dabiran et al., 2024) |
| **Perceived Value** | **Self-expression** | • This product allows me to express who I am. | (Rozenkrants et al., 2017) |

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(*continued*)

| Variables | Variable Dimensions | Items | Reference |
|---|---|---|---|
| | | • This product reflects my personality, style, and taste.<br>• Using this product helps me communicate my values and beliefs to others. | |
| | **Social Signalling** | • Owning this product helps me fit in a specific social group or community I want to associate with.<br>• This product communicates something about my social identity or status to others. | (Solomon, 1983) |
| | **Emotional Attachment** | • I have a strong emotional attachment to this product.<br>• Using this product is an emotional experience for me. | (Thomson et al., 2005) |

*Note: *reversed items.*

## Data availability

Data will be made available on request.

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